

Pressure-volume loops and arrhythmia mechanisms

Reading the pressure-volume loop

- Filling adds volume at low pressure
- Isovolumetric contraction raises pressure alone
- Ejection reduces volume
- Isovolumetric relaxation at constant end-systolic volume
- Width is stroke volume
- Enclosed area approximates stroke work
- End-diastolic relation reflects passive compliance
- End-systolic relation reflects contractile state

Loading changes the loop

- More preload widens it through Frank-Starling
- More afterload raises end-systolic volume
- More contractility lowers end-systolic volume
- Mitral regurgitation: volume escapes to a low-pressure atrium
- Aortic stenosis: pressure work and wall stress rise

Wall stress and remodelling

- Rises with pressure and radius, falls with thickness
- Pressure overload thickens the wall
- Volume overload adds sarcomeres in series
- Adaptation fails with fibrosis and capillary mismatch
- Hypertrophy is not stronger effective pumping
- Preserved ejection fraction with high filling pressure

Diastole and ventricular interaction

- Relaxation is energy-dependent
- Ischaemia impairs diastole before systole
- A stiff ventricle depends on atrial contraction
- Shared septum inside the pericardium
 - Right dilation shifts the septum leftward
 - Tamponade and high airway pressure limit both chambers
- Positive-pressure ventilation
 - Reduces venous return when volume is low
 - Reduces left transmural afterload

Oxygen demand without occlusion

- Demand tracks wall stress, rate, contractile activity
- Dilated ventricle faces greater wall stress
- Hypertrophied ventricle has lower capillary density per mass
- Tachycardia adds demand and shortens diastole

Arrhythmia initiation

- Abnormal automaticity outside the sinoatrial node
- Triggered activity depends on a preceding beat
 - Early afterdepolarisations during prolonged repolarisation
 - Delayed afterdepolarisations after calcium overload
- Re-entry needs substrate and trigger
 - Scar, fibrosis, unequal refractoriness, accessory pathway
 - A premature impulse supplies the trigger
- Antiarrhythmics can be proarrhythmic
 - Sodium block promotes another circuit
 - Potassium block permits early afterdepolarisations
 - Nodal blockers can worsen accessory-pathway rhythms

Electrocardiography as spatial sampling

- Each lead projects the cardiac vector onto an axis
- Biphasic signal when the vector is perpendicular
- Axis deviation is a clue, not a diagnosis
- Broad Q R S: block, ventricular origin, pre-excitation, pacing, drugs
- S T and T changes are not specific to occlusion
- Corrected Q T formulae fail at rate extremes

Haemodynamic consequence and priorities

- Danger comes from rate, lost synchrony, or degeneration
- Adverse features
 - Hypotension or shock
 - Ongoing ischaemic chest discomfort
 - Acute pulmonary oedema or altered consciousness
- Then find substrate and trigger
 - Ischaemia, infection, electrolytes, endocrine, drugs
 - A transient trigger does not exclude a substrate
- Serial assessment, because loading and tone change fast